The Beer–Lambert Law

Pupil sheet

Name: _____ **Date:** _____

Safety: The box is not waterproof: keep water away from it. Goggles with dyes; taste nothing.

1. What I see on the screen

The screen shows **Illuminance I** (in lux) and **Absorbance A**; I choose the sample in the **Sample in the beam:** list and press **Capture**. **Only the illuminance is measured**; the absorbance is computed by the app.

Careful — the reference. I put the **water blank** in the beam and press **Set reference I_0 from this reading**. Below the cards I check that I_0 was *chosen by me*; if not, I take it again.

That is why I write the lux in the table, not the absorbance from the screen. From the lux I can rebuild everything; from the screen's absorbance I cannot.

I_0 **(the blank) =** _____ **lx**
Write it down *before* moving on: it does not appear in the exported file.

2. What I do on my own

First I predict, before any measuring. I draw two sketches. I add the real curve over them, in another colour, at the end of the lesson.

Transmitted light (I) against c

$A = \log_{10}(I_0/I)$ against c

The rig (measured once, then nothing moves): lamp-to-sensor distance _____ cm path length l of the cup _____ cm

I measure. The blank first. The volumes are already written in. **The first row is worked as an example** — I use it as the pattern for the rest.

c	**Stock**	**Water**	I **(lx)**	$T = I/I_0$	$A = -\log_{10} T$
example			*1000*	*0.50*	*0.301*
0 (blank)	0 mL	100 mL		1.00	0.00
0.2	20 mL	80 mL			
0.4	40 mL	60 mL			
0.6	60 mL	40 mL			
0.8	80 mL	20 mL			
1.0	100 mL	0 mL			
≈ 2	—	—			
0 (end)	0 mL	100 mL			

Check: $T = 10\%$ must give $A = 1$. If I get a negative number, I used percentages instead of a fraction.

The line. The gradient, computed on the linear points only: _____. The points leave the line from about $A \approx$ _____.

How sure are we? The app's slope: _____ ± _____, R^2 = _____. The blank at the end differs from I_0 by _____ %. Is the series good? ☐ yes ☐ no

3. The conclusion I write

I predicted that, and the data show that _____

_____

The logarithm straightens the graph because _____

_____

Our instrument is trustworthy up to A = _____, and above that, because _____

_____

Exit ticket

Why does the logarithm straighten the graph?

_____